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Sequence Bioinformatics **WS 2026/27**

Assignment 1 **Due: Wed 21-Oct-2026, 10:00**

This assignment is on the lectures of Mon 12-Oct (introduction) and Wed 14-Oct (pairwise alignment), that is, on Chapter 1 of the handout, Sections 1.1–1.2 and 1.4–1.10: scoring, global, local and overlap alignment. Exercise numbers refer to the handout.

Important: please read and follow all instructions carefully. Assignments are part of the mandatory tutorial participation. Submissions that do not follow the naming or authorship requirements will not be considered.

Setup

Install an IDE for Java programming (we recommend **IntelliJ Community Edition**) and make sure that it uses **Java 17 or newer**. Download `assignment01.zip` from Ilias. It contains the skeleton `Align_YOUR_NAME.java`, which already reads a FastA file and prints the result, and the data files used below. Do not use any bioinformatics libraries: the task is to implement the algorithms *from first principles*.

1 Global, local and overlap alignment (4 points)

Complete the three methods of `Align_YOUR_NAME.java`, so that

```
java Align_YOUR_NAME <file> <mode> <match> <mismatch> <gap>
```

reads the two sequences X and Y of the FastA file `<file>` and computes, for the match score, the mismatch score and the linear gap penalty d given, an optimal

- `global` alignment by the Needleman–Wunsch algorithm (Section 1.8),
- `local` alignment by the Smith–Waterman algorithm (Section 1.9), or
- `overlap` alignment (Section 1.10),

with traceback, and prints the optimal score and an alignment that attains it. The three variants share one recurrence; they differ in the initialization of the first row and column, in the extra 0 of the local recurrence, and in where the traceback starts and where it stops. Write the matrix filling once, and make the three differences explicit in your code.

The worked examples of the handout are your test cases: with match score 1, mismatch score -1 and $d = 1$,

- `example-global.fasta` ($X =$ `CTAATGCG`, $Y =$ `TATGG`) has global score 2 (Section 1.8),
- `example-local.fasta` ($X =$ `CTAATGCC`, $Y =$ `TATGG`) has local score 3, alignment `ATG`/`ATG` (Section 1.9),
- `example-overlap.fasta` ($X =$ `ACATATT`, $Y =$ `TATTAC`) has overlap score 4 with $d = 2$, alignment `TATT`/`TATT` (Section 1.10),

- `exercise-1-4.fasta` (X = `CAGTG`, Y = `CATTG`) has global score 7 with match score 2 (Exercise 1.4).

Where several alignments are optimal, your program may report a different one than the handout; the score must agree. Include the output of your program on the four files in your submission.

2 Two mitochondrial sequences (1 point)

The file `small.fasta` contains the first few hundred bases of the human and the mouse mitochondrial genome, `mtdna-2000.fasta` the first 2,000 bases of each (GenBank `NC_012920` and `NC_005089`). Run your program on both files in all three modes with match score 1, mismatch score -1 and $d = 1$, and report the six scores and the running times. For `mtdna-2000.fasta`: how long are the local and the overlap alignment, and which of the three alignments is the appropriate one for comparing two homologous genomes? What would be needed to align the complete genomes (16,500 bases each) with your program, and why does that get difficult?

3 Which table is which (2 points)

Exercise 1.10 of the handout: three tables for X = `CATTAG` and Y = `TTAGG`, one each from the global, the local and the overlap algorithm. Say which is which and what told you, then give for each the optimal score, the cells visited by the traceback, and the alignment.

4 All optimal alignments (1 point)

Exercise 1.5: find *every* optimal global alignment of the worked example X = `CTAATGCG` and Y = `TATGG` (match 1, mismatch -1, $d = 1$) by following each predecessor that attains the maximum wherever there is a tie. How many are there, and at which cell do they part?

5 Negative expected score (1 point)

Exercise 1.9: (a) for DNA with all four bases equally frequent, match score M and mismatch score -1, for which M is the expected score of a random pair negative, and what does Smith–Waterman return for two long unrelated sequences when $M = 4$? (b) The three-residue matrix of Section 1.4, with background frequencies 0.5, 0.3 and 0.2 for L, K and F: compute the expected score; it is positive. Why does this not contradict the requirement of Section 1.9.2?

6 Reading an alignment program (1 point)

Exercise 1.20: the Java method printed there is meant to compute the optimal global alignment score, and returns the correct answer 2 on the worked example, yet contains two errors. Find both, explain why the example exposes neither, and give for each the shortest input on which the method returns a wrong score, with the correct score. (Your own program from Task 1 is a handy way to check your answers.)

Handing in

- Upload your solution to the upload directory of your tutorial group on Ilias by **Wed 21-Oct-2026, 10:00**: your Java source file, the output of your program on the test files, and your answers to Tasks 2–6 as a PDF.
- Include your **name** both in the file names and inside each file (for example in a comment). Submissions that do not follow this rule will be ignored.

If you encounter difficulties, discuss them in your tutorial or post questions in the Ilias course forum.

Tutorials

The tutorials take place on Mondays and Wednesdays, and start in the week of **26-Oct**. The first quiz, on this assignment, is taken there (Mon 26-Oct or Wed 28-Oct, depending on your group).

Use of ChatGPT or similar systems

If you use ChatGPT or similar AI systems to help with this assignment, you must clearly document this.

- In science, authorship and attribution must always be transparent. If you did not personally author a text, code snippet, or figure, add a clear attribution statement directly next to it.
- You are responsible for understanding and being able to explain all submitted material.
- If generated code is part of your solution, describe how you verified that it is correct: the test cases of Task 1 are the minimum.
- Tutors will not debug or evaluate text or code produced by AI systems.